

Hikmah
HACKERS

Week 2: Variables

Boxes that remember things

DARUL ULOOM MICHIGAN · FRIDAYS

TODAY

Our plan for 40 minutes

Time	What we do
0–5 min	Warm-up: show your All About Me
5–15 min	The big idea: a labelled box
15–30 min	Type along: make, print and change variables
30–37 min	Your turn + bug hunt
37–40 min	Mini-project: My Profile Card

WARM-UP

Last week, we typed every fact by hand.

Problem

Change your favourite food? Find every line that says it.

```
print("I love pizza")
```

Fix

Write it once in a box. Use the box everywhere.

```
food = "pizza"
```

THE BIG IDEA

A variable is a box with a label on it.

1

Pick a label for the box.

```
name
```

2

Put something inside.

```
name = "Mohammad"
```

3

Ask for it by its label.

```
print(name)
```

LET'S CODE · TYPE ALONG

Your first variable

TYPE THIS, THEN PRESS RUN

```
name = "Mohammad"
print(name)
```

YOU SHOULD SEE

```
Mohammad
```

Read = as “put into”, not “equals”. Put "Mohammad" into the box called name.

QUOTES MATTER

Quotes or no quotes?

With quotes

```
print("name")
```

Prints the exact letters: name

No quotes

```
print(name)
```

Looks inside the box called name: Mohammad

TEXT VS NUMBERS

What's in the box?

Code	What's inside	Type
`food = "biryani"`	the letters b-i-r-y-a-n-i	text (a string)
`grade = 8`	the number 8	number (an integer)
`age = "13"`	the characters 1 and 3	text! Quotes make it text

TEXT VS NUMBERS

Numbers can do math. Text can't.

TYPE THIS, THEN PRESS RUN

```
grade = 8
print(grade + 1)
print("8" + "1")
```

YOU SHOULD SEE

```
9
81
```

"8" + "1" just sticks the text together. That's why you get 81, not 9.

LET'S CODE · TYPE ALONG

Printing variables with text

TYPE THIS, THEN PRESS RUN

```
name = "Mohammad"
grade = 9
print("Salaam, I am", name)
print("I am in grade", grade)
```

YOU SHOULD SEE

```
Salaam, I am Mohammad
I am in grade 9
```

Use commas inside print(). Python puts a space between each part for you.

LET'S CODE · TYPE ALONG

Boxes can change

TYPE THIS, THEN PRESS RUN

```
prayer = "Fajr"
print("Next prayer:", prayer)

prayer = "Dhuhr"
print("Next prayer:", prayer)
```

YOU SHOULD SEE

```
Next prayer: Fajr
Next prayer: Dhuhr
```

A new value replaces the old one. Python runs top to bottom, so order matters.

NAMING RULES

Good names, bad names

Works	Breaks	Why
`favourite_food`	`favourite food`	No spaces. Use _ instead
`player2`	`2player`	Can't start with a number
`name`	`print`	Don't reuse Python's own words

Python cares about capitals: Name and name are two different boxes.

YOUR TURN · 5 MIN

Your family's favourite meal

```
meal = "kabsa"
cook = "my mom"
day = "Friday"
print("On", day, cook, "makes",
meal)
```

Finished early?

Change just one variable and press Run. One line changed, but the sentence updated. That's the superpower.

BUG HUNT

Four mistakes everyone makes

Broken	What happens	Fix
`print(nmae)`	NameError: no box called nmae	Check the spelling
`print("name")`	Prints the word name	Remove the quotes
`print(score) then score = 10`	NameError: box doesn't exist yet	Create it first
`print("Grade " + 8)`	TypeError: text + number	Use , instead of +

MINI-PROJECT · DUE NEXT FRIDAY

My Profile Card

Required

- At least 5 variables (name, grade, favourite food, hobby, dream project…)
- Print a card that uses every variable at least once
- Add a border or title line so it looks like a card
- Change one variable, press Run again, and check the card updates

Bonus

```
grade = 8
print("Years left:", 12 - grade)
```

Add a number variable and do math with it.

Turn it in at hikmah-hackers.github.io/submit · due Friday, Nov 13

NEXT FRIDAY

input(): let the user fill the box

```
name = input("What is your name? ")
print("Salaam,", name)
```

Next week your programs start talking back.